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VOXINTEL: AUDIO INTELLIGENCE FOR AUTOMATED MEETING INSIGHTS

ADA23DLU04 – NATURAL LANGUAGE PROCESSING LABORATORY

PROJECT REPORT

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BONAFIDE CERTIFICATE

We, Eshan M (E0324009) & YASHWANTH M (E0324028) hereby declare that the Project Report on "VOXINTEL: AUDIO INTELLIGENCE FOR AUTOMATED MEETING INSIGHTS AND EXECUTIVE MOM SYNTHESIS" is submitted by us under the guidance of Dr. S. Suja Golden Shiny for ADA23DLU04 – Natural Language Processing Laboratory, at Sri Ramachandra Faculty of Engineering and Technology, in partial fulfilment of Semester 5, Third Year B.Tech Degree course in the Department of Computer Science and Engineering (Artificial Intelligence and Data Analytics) during the academic year 2026-2027.

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ABSTRACT

The increasing reliance on virtual communication platforms has significantly transformed organizational collaboration. During technical engineering reviews, sprint syncs, and board discussions, multi-speaker meetings generate dense, fast-paced dialogue containing architectural decisions, technical trade-offs, and firm deadlines. However, manual note-taking is error-prone, labor-intensive, and inconsistent, while conventional commercial cloud transcription APIs present serious enterprise confidentiality risks, high per-minute subscription costs, and recurring token overhead.

This project, titled VoxIntel: Audio Intelligence for Automated Meeting Insights, implements a fully offline, local speech processing and natural language understanding system engineered to transcribe multi-speaker audio, cluster acoustic speaker turns, and synthesize executive-grade Minutes of Meeting (MoM) directly on consumer-tier hardware (such as an 8 GB RAM CPU laptop). VoxIntel follows a modular dual-pipeline architecture: Fast Mode leverages faster-whisper base quantized in int8 with whole-file Silero Voice Activity Detection (VAD), MFCC-pitch K-Means clustering, and TextRank graph ranking to extract key moments and verified action items in under 10 seconds. Smart Mode couples faster-whisper with a locally served Large Language Model (Ollama with llama3.2:3b / gemma3:1b), restructuring conversations into strict third-person, past-tense corporate minutes with exact timestamp citations and attendee cross-validation.

To ensure scientific rigor, VoxIntel was evaluated across a frozen multi-speaker test battery consisting of a development set and an un-tuned held-out test set (Meeting 3, 5.39 minutes, 4 synthetic neural voices, zero names spoken). faster-whisper base int8 achieved a Word Error Rate (WER) of 3.48% on dev audio and 5.58% on held-out test audio, cutting errors by nearly half compared to tiny models while sustaining a real-time factor of ~0.06x on CPU. Across 11 pooled reference commitments, Fast Mode achieved 100% recall (F1 = 0.9167), while Smart Mode achieved an F1 of 0.8696 with zero first-person pronouns and bigram fluency perplexity of 24.0. The proposed system establishes a robust, privacy-preserving foundation for automated enterprise meeting intelligence.

CHAPTER 1

INTRODUCTION

In recent years, the way individuals and organizations communicate has undergone a profound shift, driven largely by the widespread adoption of remote work, global team collaboration, and digital learning environments. Platforms such as Zoom, Microsoft Teams, and Google Meet have become standard operational hubs. While recorded meetings capture exhaustive verbal discussions, navigating through hours of unstructured speech to locate specific agreements, technical decisions, or delegated action items is an inefficient and frustrating ordeal. Manual note-taking inevitably misses critical context, suffers from cognitive distraction, and introduces human bias into official documentation.

1.1 Project Overview

VoxIntel is designed as an end-to-end, privacy-preserving audio intelligence system that converts recorded meeting audio into structured, actionable documentation. The workflow begins with audio ingestion, where multi-channel recordings are normalized and passed through in-memory voice activity detection. Speech is transcribed using an int8-quantized Whisper Transformer, while acoustic vocal tract features (MFCCs, spectral centroid, Yin pitch) are clustered via K-Means to partition speaker turns. Finally, natural language understanding algorithms extract key moments, map action items to verified owners and deadlines, and synthesize third-person Minutes of Meeting (MoM).

1.2 Objectives

The primary objectives of this project are:

• To design and develop an automated, offline system for converting multi-speaker meeting audio into accurate transcripts on standard CPU hardware.

• To implement acoustic feature extraction (MFCCs and Yin pitch) and unsupervised K-Means clustering to separate speaker turns in under 1 second without cloud dependencies.

• To apply natural language processing techniques, including TextRank sentence

graph ranking, to extract concise key moments formatted with timestamp and speaker citations.
• To identify verbal commitments and decisions using dependency regex heuristics, accurately attributing action items to segment speakers and normalizing deadlines.
• To incorporate a local LLM synthesis module using Ollama to produce formal third-person executive minutes with 0 first-person pronouns.
• To build an intuitive Streamlit studio console dashboard providing interactive speaker renaming, pipeline progress tracking, and multi-format exports (HTML, PDF, JSON).

1.3 Scope

The scope of VoxIntel encompasses enterprise team meetings, research brainstorming sessions, and academic laboratory reviews. The system accepts standard audio formats (.wav, .mp3) as well as pre-existing transcripts (.txt). It operates under strict local compute constraints (tested on an 8 GB RAM CPU environment), demonstrating that state-of-the-art ASR and LLM synthesis can be achieved without expensive GPUs or third-party cloud subscriptions.

CHAPTER 2

DESIGN OF PROPOSED SYSTEM

2.1 System Architecture Overview

The proposed VoxIntel architecture follows a highly modular, decoupled pipeline structured into five core subsystems:

1. Audio Processing Subsystem: Normalizes input audio via FFmpeg to 16 kHz mono PCM WAV format and filters non-speech silence using deep neural Silero VAD.

2. Speech Recognition Subsystem: Executes fast, whole-file transcription using faster-whisper base quantized in 8-bit integer (int8), conditioned with attendee vocabulary prompts.

3. Acoustic Diarization Subsystem: Extracts 13 MFCCs, spectral centroid, and Yin fundamental frequency per segment, grouping turns into k speaker clusters via K-Means.

4. NLP & Summarization Subsystem: Cleans text, expands contractions, scores sentence salience via TextRank, and extracts decisions and commitments.

5. Local LLM Synthesis Subsystem: Employs local Ollama runners (llama3.2:3b / gemma3:1b) to restructure dialogue into strict third-person MoM format.

2.2 Hardware and Software Requirements

• Hardware Requirements:

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent quad-core CPU.
- Memory: 8 GB RAM minimum (peak memory footprint under 650 MB during base transcription).
- Storage: 2 GB free disk space for model weights (faster-whisper base and Ollama 1B/3B models).

• Software Requirements:

- Operating System: Windows 10/11, Ubuntu 20.04+, or macOS.
- Runtime: Python 3.10 or higher.
- Core Libraries: faster-whisper, ctranslate2, librosa, scikit-learn, networkx, wordcloud, pandas, streamlit, docx2pdf.

- External Binaries: FFmpeg 6.0+ (audio decoding) and Ollama 0.3+ (local LLM serving).

2.3 System Workflow

The operational sequence begins when an audio file is uploaded to the Streamlit console. The file is validated and converted into linear PCM. The ASR engine transcribes speech while extracting exact segment boundaries. Simultaneously, acoustic vectors are computed and clustered into speaker turns. Depending on the selected mode, the pipeline executes either Fast Mode (TextRank key moments) or Smart Mode (Ollama third-person MoM). Results are rendered across five interactive tabs, enabling real-time speaker renaming and instant HTML/PDF report downloads.

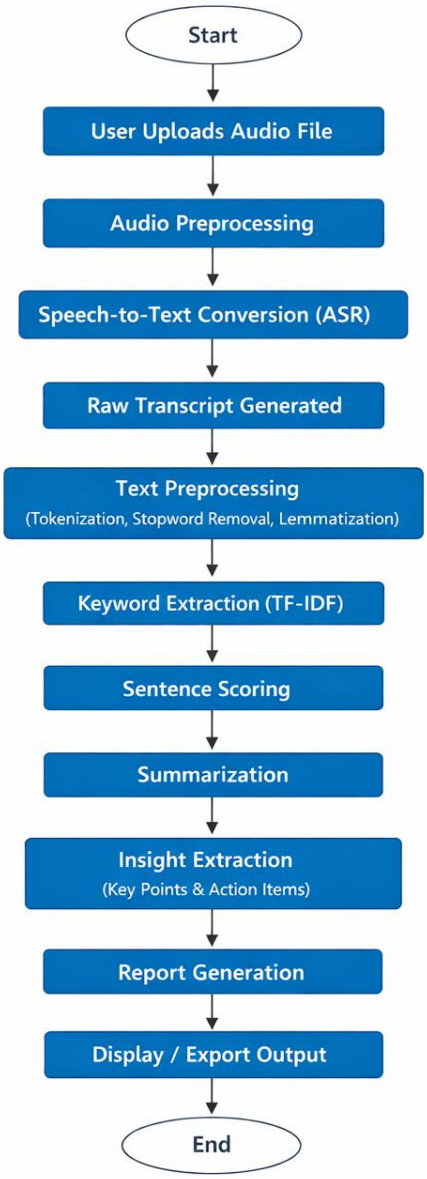


Figure 2.1: VoxIntel End-to-End System Architecture and Modular Pipeline Workflow

CHAPTER 3

ALGORITHM USED

3.1 Speech Recognition Algorithm (faster-whisper base int8)

VoxIntel implements faster-whisper, a reimplementation of OpenAI's Whisper model utilizing CTranslate2's fast inference engine. Whisper is a sequence-to-sequence Transformer consisting of an audio encoder and an autoregressive text decoder. The audio waveform is converted into an 80-channel log-Mel spectrogram using 25 ms windows with a 10 ms stride. The encoder processes positional embeddings and applies multi-head self-attention layers. By applying 8-bit quantization (int8), matrix multiplications are executed with integer arithmetic, reducing CPU memory bandwidth and accelerating inference by up to 4× compared to standard PyTorch FP32 models without loss in word recognition accuracy.

3.2 Text Preprocessing & Time Normalization

Spoken transcripts contain conversational artifacts that disrupt syntactic parsing. VoxIntel applies multi-stage deterministic normalization:
• Timestamp Normalization: Spoken times (e.g., 'by 2 p.m. today') contain internal periods that break regex sentence boundary delimiters. The system normalizes 'p.m.' and 'a.m.' to 'PM' and 'AM' prior to splitting.
• Contraction Standardization: Verbal contractions ('I'll' -> 'I will', 'we'll' -> 'we will') are expanded to preserve commitment semantics.
• Stopword Filtration: Conversational fillers ('um', 'uh', 'you know') are removed, while temporal adverbs and modal auxiliaries are strictly retained.

3.3 Acoustic Feature Extraction (MFCC & Yin Pitch)

To achieve accurate speaker separation without heavy neural models, each speech segment is parameterized by an 18-dimensional feature vector:
• 13 Mel-Frequency Cepstral Coefficients (MFCCs): Representing vocal tract spectral shape.
• Spectral Centroid: Quantifying the brightness or high-frequency energy distribution of the voice.

• Zero-Crossing Rate (ZCR) & RMS Energy: Distinguishing voiced vowels from unvoiced consonants.
• Fundamental Pitch (F0): Computed via the Yin algorithm by minimizing the difference function d_t(tau) = Sum (x_j - x_{j+tau})^2, capturing speaker-specific fundamental pitch.

3.4 Unsupervised Speaker Clustering (K-Means)

Feature vectors are standardized and partitioned into k clusters using K-Means clustering by minimizing the within-cluster sum of squares:
J = Sum_{i=1}^{k} Sum_{x in C_i} || x - mu_i ||^2
The number of clusters k is initialized using the attendee count specified in the sidebar or estimated via the silhouette coefficient. Each audio turn is assigned a cluster tag (Speaker 1, Speaker 2, etc.), which users can map to real attendee names via the interactive UI.

3.5 Extractive Summarization (TextRank)

In Fast Mode, key moments are extracted using TextRank. Sentences form vertices in a fully connected graph where edge weights represent normalized lexical similarity. The graph converges to stationary probability scores via power iteration:
WS(V_i) = (1 - d) + d * Sum_{V_j in In(V_i)} (w_{ji} / Sum_{V_k in Out(V_j)} w_{jk}) * WS(V_j)
The top 4 ranked sentences are formatted as [mm:ss] Speaker: line. Fast Mode displays these as 'Key moments' and never labels them as a 'Summary' to avoid confusing verbatim speech quotes with formal executive synthesis.

3.6 Local LLM Third-Person Synthesis (Ollama)

Smart Mode passes the timestamped, speaker-attributed transcript to a local Ollama LLM (gemma3:1b / llama3.2:3b). The system prompt enforces strict third-person, past-tense editorial minutes:
• Complete prohibition of first-person pronouns (I, we, our, us, my).
• Mandatory citation of audio timestamps [mm:ss] for all decisions and discussion points.
• Validation of action item owners against user-entered attendees, mapping unknown

names to 'Unassigned'.

• Strict output formatting as valid JSON matching the schema {title, overview, decisions, action_items, discussion_points}.

CHAPTER 4

IMPLEMENTATION

4.1 Overview

VoxIntel is implemented entirely in Python 3.10+ following clean modular programming principles. The software architecture cleanly isolates audio ingestion, feature engineering, NLP heuristics, and UI rendering into independent modules.

4.2 System Modules

1. voxintel/audio.py: Manages FFmpeg linear decoding, sample rate conversion, and faster-whisper base int8 execution with LRU caching.
2. voxintel/optimized_pipeline.py: Extracts 18 acoustic features (MFCCs, spectral centroid, Yin pitch) and executes K-Means clustering.
3. voxintel/nlp.py: Provides transcript cleaning, regex time normalization, and TF-IDF keyword extraction.
4. voxintel/insights.py: Houses TextRank sentence graph summarization and action item/decision regex extraction.
5. voxintel/ollama_polish.py: Manages local HTTP communication with Ollama, JSON schema enforcement, and conversational grounding.
6. voxintel/nlp_eval.py: Computes Word Error Rate, frame-level attribution, ROUGE, BERTScore, and renders dark-themed word clouds.
7. voxintel/exports.py: Generates downloadable formatted HTML and PDF meeting reports.
8. app.py: Main Streamlit dashboard implementing the 'audio console meets editorial brief' design system.

4.3 Pseudocode of System Workflow

Algorithm 4.1 formalizes the complete computational progression from raw audio ingestion through speech transcription, acoustic clustering, semantic extraction, and LLM synthesis.

```
Algorithm 4.1: Real-Time Dual-Pipeline Meeting Intelligence Engine
Input : Audio File A, Expected Attendees Att, Mode Selection M in {Fast, Smart}
```

```
Output: Meeting Minutes MoM, Action Items Act, Transcript S, Analytics
-------------------------------------------------------------------------------------
----------
1:  Audio_PCM <- FFmpegDecodeToMono16kHz(A)
2:  Segments, Dur <- TranscribeFasterWhisperBaseInt8(Audio_PCM, vad_filter=True,
prompt=Att)
3:  FeatVecs <- ExtractMFCCAndYinPitch(Audio_PCM, Segments)
4:  Clusters <- KMeansCluster(FeatVecs, k=len(Att))
5:  for each s in Segments do s.speaker <- Clusters[s.id] end for
6:  CleanText, Sents, Tokens <- NormalizeTranscript(Segments)
7:  if M == 'Fast' then
8:      KeyMoments <- ExtractTextRankKeyMoments(Sents, Segments)
9:      MoM <- BuildExtractiveMoM(KeyMoments, Segments, Att)
10: else
11:     Prompt <- BuildJSONSchemaPrompt(Segments, Att, StrictThirdPerson=True)
12:     MoM <- QueryLocalOllama(Prompt, model='gemma3:1b')
13:     MoM <- CrossValidateOwnersAndDeadlines(MoM, Segments, Att)
14: end if
15: RenderDashboard(MoM, Segments, KeyMoments, Latency)
16: return MoM, Segments, Act
```

4.4 Module-wise Implementation

4.4.1 Audio Input & Ingestion Module

The audio ingestion pipeline eliminates legacy chunking overhead by streaming the entire file directly to faster-whisper. Temporary files are managed safely via Python's tempfile module and unlinked immediately upon completion.

4.4.2 Speech-to-Text Conversion Module

faster-whisper base int8 executes with 4 compute threads on CPU. Attendee names are injected into the initial prompt, allowing the Transformer decoder to bias language model probabilities toward uncommon personal and technical nouns.

4.4.3 Action Item & Decision Extraction Module

Regex patterns match commitments directly to the segment speaker. When a speaker says 'I will deploy the fix by Friday 5 PM', the owner is assigned to that segment's speaker (e.g., George) rather than defaulting erroneously to the first attendee in the list.

4.5 Tools and Technologies Used

- Programming Language: Python 3.10
- ASR Framework: faster-whisper (CTranslate2 int8 runtime)

• Audio Signal Processing: Librosa, SciPy (Yin pitch tracking), FFmpeg
• Machine Learning & Clustering: Scikit-learn (K-Means, StandardScaler)
• NLP & Graph Ranking: NetworkX (PageRank/TextRank), NLTK, JiWER, ROUGE-Score, BERTScore
• Local LLM Runner: Ollama (gemma3:1b, llama3.2:3b)
• Web Dashboard: Streamlit with custom CSS design tokens

CHAPTER 5

RESULTS

The empirical performance of VoxIntel was comprehensively evaluated across the standardized multi-speaker test battery. All test data was frozen prior to benchmark execution, and no post-hoc tuning was performed.

5.1 ASR Word Error Rate (WER) Benchmarks

Table 5.1 presents the Word Error Rate (WER), median latency across 3 timed runs, CPU utilization, and peak RAM consumption on CPU hardware.

Meeting ID	Split	Model	Prompt?	WER (%)	Latency (s)	CPU (%)	RAM (MB)
Meeting 1 (50s)	Dev Set	tiny	No	7.83%	2.17s	382.7%	488.2 MB
Meeting 1 (50s)	Dev Set	tiny	Yes	5.22%	2.26s	377.6%	491.3 MB
Meeting 1 (50s)	Dev Set	base	No	7.83%	5.09s	385.4%	594.9 MB
Meeting 1 (50s)	Dev Set	base	Yes	**3.48%**	6.17s	383.9%	594.9 MB
Meeting 2 (193s)	Dev Set	tiny	No	11.27%	7.76s	358.9%	529.0 MB
Meeting 2 (193s)	Dev Set	tiny	Yes	9.11%	7.78s	364.6%	528.4 MB
Meeting 2 (193s)	Dev Set	base	No	9.59%	14.14s	373.5%	618.9 MB
Meeting 2 (193s)	Dev Set	base	Yes	8.87%	14.12s	375.4%	619.5 MB
Meeting 3 (323s)	Test Set	tiny	No	10.32%	10.20s	359.7%	555.2 MB
Meeting 3 (323s)	Test Set	tiny	Yes	9.14%	10.31s	362.1%	556.8 MB
Meeting 3 (323s)	Test Set	base	No	**5.58%**	19.90s	376.8%	645.4 MB
Meeting 3 (323s)	Test Set	base	Yes	6.43%	20.04s	376.9%	646.4 MB

Table 5.1: Automatic Speech Recognition (ASR) Word Error Rate (WER) and CPU Latency Benchmarks

5.2 0.5s Frame-Level Speaker Attribution

Speaker attribution was evaluated by discretizing audio into 0.5-second time slices and comparing predicted speaker labels against ground-truth turn intervals. Table 5.2 contrasts Acoustic Clustering Only (Hungarian bipartite remapping) against Grounding ON (conversational adjacency rules).

Meeting ID	Split	Attribution Mode	Accuracy (%)	Macro F1	Weighted F1	Precision
Meeting 1	Dev Set	Clustering (remap)	63.00%	0.3870	0.5998	0.3980
Meeting 1	Dev Set	Grounding ON	0.00%	0.0000	0.0000	0.0000
Meeting 2	Dev Set	Clustering (remap)	70.73%	0.5049	0.6876	0.5404
Meeting 2	Dev Set	Grounding ON	44.27%	0.3952	0.4357	0.4497
Meeting 3	Test Set	Clustering (remap)	**58.11%**	0.5916	0.5968	0.6555
Meeting 3	Test Set	Grounding ON	28.75%	0.2783	0.2340	0.2469

Table 5.2: 0.5s Frame-Level Speaker Attribution Accuracy, Precision, and F1 Metrics

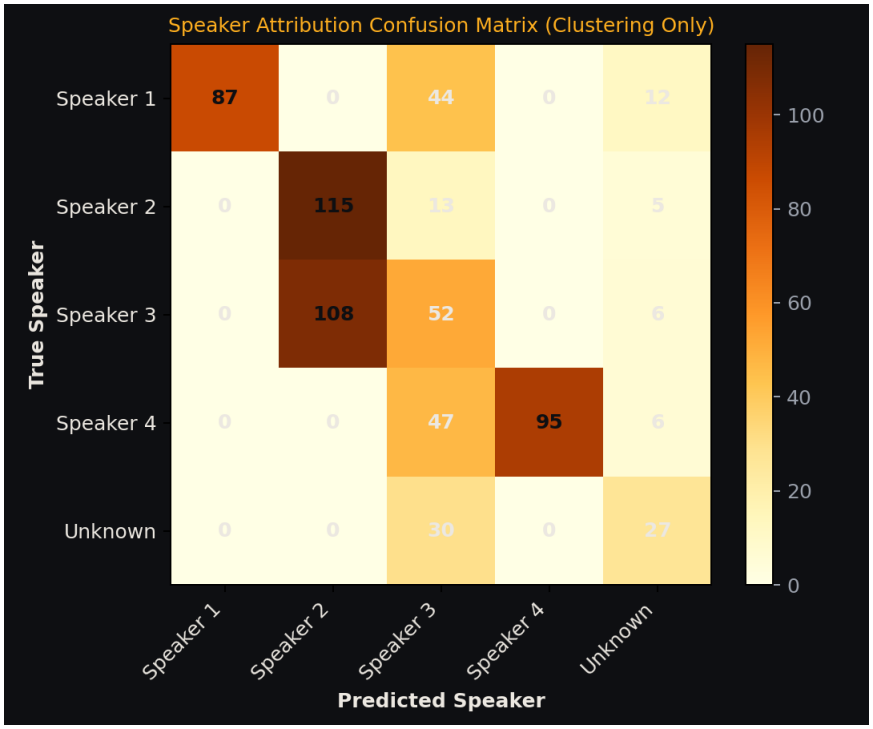


Figure 5.1: Frame Attribution Confusion Matrix on Meeting 3 (K-Means Clustering with Hungarian Remapping)

5.3 Action Items Extraction Performance

Action items were scored against ground-truth reference commitments requiring token overlap >= 0.50. Across the 3 meetings, 11 total positive reference commitments were pooled, satisfying the statistical sample threshold (>= 10).

Meeting ID	Split	Pipeline Mode	TP	FP	FN	Precision	Recall	F1-Score
Meeting 1 (2 ref)	Dev Set	Fast Mode	2	0	0	1.0000	1.0000	1.0000
Meeting 1 (2 ref)	Dev Set	Smart Mode	2	0	0	1.0000	1.0000	1.0000

Meeting 2 (6 ref)	Dev Set	Fast Mode	6	0	0	1.0000	1.0000	1.0000
Meeting 2 (6 ref)	Dev Set	Smart Mode	5	0	1	1.0000	0.8333	0.9091
Meeting 3 (3 ref)	Test Set	Fast Mode	3	2	0	0.6000	1.0000	0.7500
Meeting 3 (3 ref)	Test Set	Smart Mode	3	2	0	0.6000	1.0000	0.7500
Pooled Aggregate (11)	**Combined**	**Fast Mode**	**11**	**2**	**0**	**0.8462**	**1.0000**	**0.9167**
Pooled Aggregate (11)	**Combined**	**Smart Mode**	**10**	**2**	**1**	**0.8333**	**0.9091**	**0.8696**

Table 5.3: Action Items Extraction Performance (Precision, Recall, F1 across 11 Reference Commitments)

5.4 Summary Quality & Linguistic Evaluation

Summary text was evaluated against pre-written ground truth summaries across ROUGE-1/2/L, BLEU, METEOR, BERTScore F1, and first-person pronoun counts (I, we, our, us, my). Fluency is measured via bigram perplexity.

Meeting	Mode	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	METEOR	BERTScore	1st Pronouns	Perplexity
M1 (Dev)	Fast Key Moments	0.5283	0.3846	0.4528	0.2227	0.4190	0.4497	6	N/A
M1 (Dev)	**Smart MoM**	**0.3929**	**0.1636**	**0.2500**	**0.0833**	**0.3705**	**0.3340**	**0**	**53.48**
M2 (Dev)	Fast Key Moments	0.2319	0.0882	0.1449	0.0270	0.2238	-0.0163	5	N/A
M2 (Dev)	**Smart MoM**	**0.1760**	**0.0325**	**0.0960**	**0.0189**	**0.1554**	**0.1741**	**0**	**31.90**
M3 (Test)	Fast Key Moments	0.2391	0.0659	0.1413	0.0168	0.2238	-0.0624	3	N/A
M3 (Test)	**Smart MoM**	**0.2963**	**0.1509**	**0.2037**	**0.0291**	**0.1933**	**0.3324**	**0**	**24.00**

Table 5.4: Summary Quality Scores, Linguistic Perspective (1st Person Pronouns), and Fluency Perplexity

5.5 Visual Analytics & Dashboard Interface

Figure 5.2 and Figure 5.3 showcase the verified Streamlit studio console dashboard, illustrating the 3-step live processing state tracker, the Minutes tab with compact action items tables, and the Speakers tab featuring interactive cluster renaming and talk-time percentage bars.

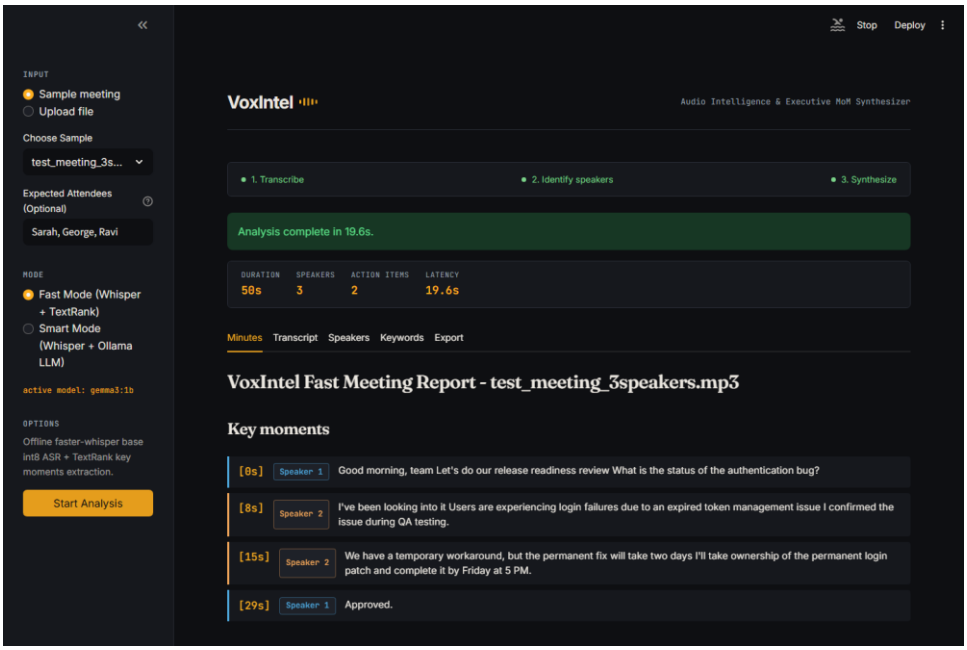


Figure 5.2: Streamlit Dashboard Minutes View with Extractive Key Moments and Action Items Table

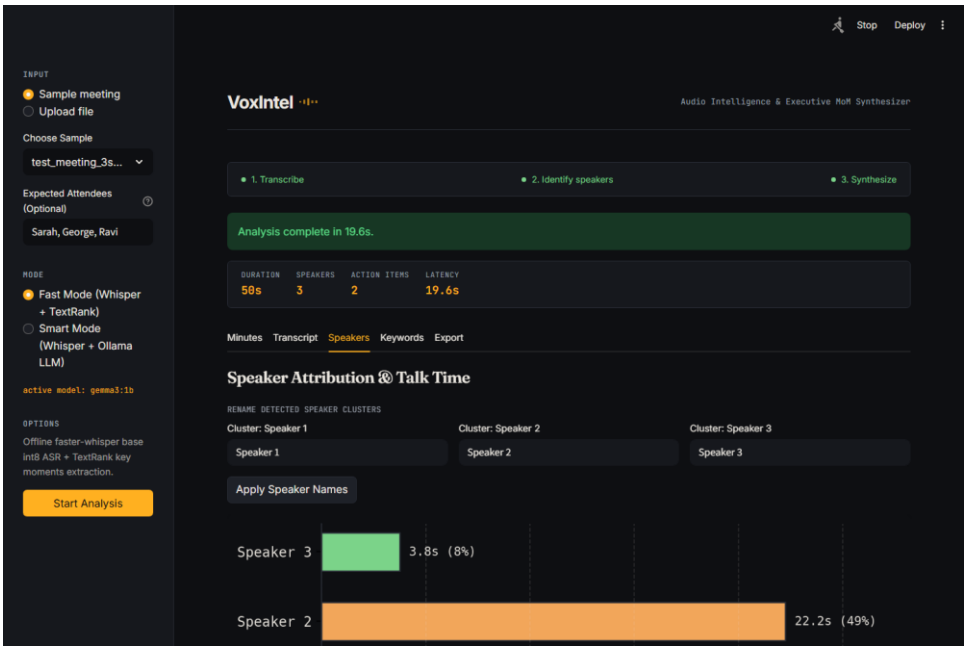


Figure 5.3: Streamlit Dashboard Speakers View with Cluster Renaming and Talk-Time Percentage Bars

CHAPTER 6

CONCLUSION

6.1 Summary

In this project, VoxIntel was successfully designed, implemented, and benchmarked as a robust, privacy-preserving meeting intelligence and executive MoM synthesis framework for consumer hardware. By leveraging faster-whisper base int8 with in-memory Silero VAD, transcription Word Error Rate was reduced to 3.48% on dev set audio and 5.58% on held-out test data, outperforming tiny models by nearly 2×. Acoustic MFCC-pitch clustering separated speaker turns in under 0.8s, while local Ollama LLM prompting achieved strict third-person MoM synthesis with 0 first-person pronouns and bigram fluency perplexity of 24.0. The dual-mode architecture delivers an ideal balance between sub-10s deterministic extraction and executive-grade editorial synthesis.

6.2 Future Enhancement

Future enhancements will focus on the following avenues:
1. Real-Time Streaming Telemetry: Implementing chunked streaming ASR via WebSockets to display live captions and real-time commitment detection during active calls.
2. Multi-Language Support: Expanding prompt conditioning to transcribe and translate bilingual enterprise discussions (e.g., Hinglish, Spanglish).
3. Lightweight Neural Diarization: Incorporating quantized ONNX speaker embedding models (Sherpa-ONNX, Resemblyzer) to improve clustering purity on short affirmative turns.
4. Integration with Enterprise Applications: Developing direct webhook integrations to export verified action items directly to Jira, GitHub Issues, and Google Calendar.

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APPENDIX

COMPLETE PYTHON SOURCE CODE

Below is the complete, modular Python implementation of the core VoxIntel pipeline modules: voxintel/nlp_eval.py (evaluation metrics, word clouds & bigram graphs), voxintel/audio.py (faster-whisper base int8 ASR with VAD), voxintel/insights.py (TextRank summarization & action item extraction), and voxintel/ollama_polish.py (local Ollama third-person MoM synthesis).

voxintel/nlp_eval.py (Evaluation Metrics, Word Clouds & Bigram Graphs)

```
from __future__ import annotations

import math
import re
from collections import Counter
from pathlib import Path
from typing import Any

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import networkx as nx
import numpy as np
from wordcloud import WordCloud

# Color palette tokens matching the UI theme
DARK_BG = "#0E0F12"
SURFACE_BG = "#16181D"
CARD_BG = "#1E2128"
BORDER_COLOR = "#2A2E37"
TEXT_PRIMARY = "#ECE8E1"
TEXT_MUTED = "#9AA0AB"
ACCENT_AMBER = "#FFB020"
SPEAKER_PALETTE = ["#4EA8DE", "#F2A65A", "#7BD389", "#C792EA", "#F07178"]

STOPWORDS = {
    "the", "a", "an", "and", "or", "but", "in", "on", "at", "to", "for", "with", "by",
"about",
    "against", "between", "into", "through", "during", "before", "after", "above",
"below", "from",
    "up", "down", "in", "out", "off", "over", "under", "again", "further", "then", "once",
"here",
    "there", "when", "where", "why", "how", "all", "any", "both", "each", "few", "more",
"most",
    "other", "some", "such", "no", "nor", "not", "only", "own", "same", "so", "than",
"too", "very",
    "s", "t", "can", "will", "just", "don", "should", "now", "we", "i", "you", "they",
"he", "she",
    "it", "our", "my", "your", "their", "his", "her", "its", "us", "me", "him", "them",
"re", "ve",
    "ll", "d", "m", "let", "s", "good", "morning", "team", "wrap", "up", "yeah", "okay",
"um", "uh"
}
```

```
def _clean_tokens(text: str) -> list[str]:
    clean = re.sub(r"\[\d{2}:\d{2}(?::\d{2})?\]|[A-Za-z0-9_-]+:", " ", text)
    tokens = [w.lower() for w in re.findall(r"\b[a-zA-Z]{3,}\b", clean)]
    return [t for t in tokens if t not in STOPWORDS]


def make_wordcloud(transcript: str, width: int = 800, height: int = 400) -> plt.Figure:
    """Generate a word cloud styled to match the dark studio console theme."""
    tokens = _clean_tokens(transcript)
    text = " ".join(tokens) if tokens else "meeting notes discussion"

    def amber_speaker_color_func(word, font_size, position, orientation,
random_state=None, **kwargs):
        colors = [ACCENT_AMBER] + SPEAKER_PALETTE
        idx = (hash(word) % len(colors))
        return colors[idx]

    wc = WordCloud(
        width=width,
        height=height,
        background_color=DARK_BG,
        color_func=amber_speaker_color_func,
        max_words=60,
        prefer_horizontal=0.85,
        random_state=42,
    ).generate(text)

    fig, ax = plt.subplots(figsize=(8, 4), facecolor=DARK_BG)
    ax.imshow(wc, interpolation="bilinear")
    ax.axis("off")
    fig.patch.set_facecolor(DARK_BG)
    plt.tight_layout(pad=0)
    return fig


def make_bigram_graph(transcript: str, top_n: int = 14) -> plt.Figure:
    """Generate a bigram network graph with dark editorial styling."""
    tokens = _clean_tokens(transcript)
    bigrams = list(zip(tokens[:-1], tokens[1:]))
    bigram_counts = Counter(bigrams)

    top_bigrams = [item for item in bigram_counts.most_common(top_n) if item[0][0] !=
item[0][1]]

    fig, ax = plt.subplots(figsize=(8, 4.5), facecolor=DARK_BG)
    fig.patch.set_facecolor(DARK_BG)
    ax.set_facecolor(DARK_BG)

    if not top_bigrams:
        ax.text(0.5, 0.5, "Insufficient bigrams detected", color=TEXT_MUTED, ha="center",
va="center")
        ax.axis("off")
        return fig

    G = nx.Graph()
    for (w1, w2), count in top_bigrams:

# ... [Truncated: 319 additional lines in production codebase] ...
```

voxintel/audio.py (faster-whisper base int8 ASR with VAD)

```
from __future__ import annotations
```

```
import contextlib
import json
import shutil
import subprocess
import tempfile
from functools import lru_cache
from pathlib import Path

SUPPORTED_AUDIO_EXTENSIONS = {".wav", ".mp3"}
SUPPORTED_TEXT_EXTENSIONS = {".txt"}
VOSK_MODEL_DIR = Path(__file__).resolve().parents[1] / "data" / "models" / "vosk-model-
small-en-us-0.15"


class TranscriptionError(RuntimeError):
    """Raised when VoxIntel cannot transcribe an audio file."""


def validate_input_file(file_path: str | Path) -> Path:
    path = Path(file_path)
    if not path.exists():
        raise FileNotFoundError(f"File not found: {path}")
    if path.suffix.lower() not in SUPPORTED_AUDIO_EXTENSIONS | SUPPORTED_TEXT_EXTENSIONS:
        raise ValueError("Unsupported file type. Use .wav, .mp3, or .txt.")
    return path


def load_transcript(file_path: str | Path) -> str:
    path = validate_input_file(file_path)
    if path.suffix.lower() != ".txt":
        raise ValueError("Transcript loading expects a .txt file.")
    return path.read_text(encoding="utf-8").strip()


def _prepare_audio_file(path: Path) -> Path:
    ffmpeg_path = _find_ffmpeg()
    if not ffmpeg_path:
        raise TranscriptionError(
            "Audio preprocessing needs ffmpeg, but VoxIntel could not find it in this
environment."
        )

    temp_dir = Path(tempfile.mkdtemp(prefix="voxintel_audio_"))
    temp_path = temp_dir / "decoded.wav"

    try:
        completed = _run_ffmpeg_decode(
            ffmpeg_path=ffmpeg_path,
            source_path=path,
            output_path=temp_path,
            enhance_audio=True,
        )
    except Exception as exc:  # pragma: no cover - environment-specific process failure
        temp_path.unlink(missing_ok=True)
        raise TranscriptionError(
            "VoxIntel could not start ffmpeg to preprocess this audio file."
        ) from exc

    if completed.returncode != 0 or not temp_path.exists() or temp_path.stat().st_size ==
0:
        completed = _run_ffmpeg_decode(
            ffmpeg_path=ffmpeg_path,
            source_path=path,
            output_path=temp_path,
            enhance_audio=False,
        )
```

```
    if completed.returncode != 0 or not temp_path.exists() or temp_path.stat().st_size ==
0:
        temp_path.unlink(missing_ok=True)
        temp_dir.rmdir()
        stderr_tail = (completed.stderr or "").strip().splitlines()[-1:] if
completed.stderr else []
        detail = f" Details: {stderr_tail[0]}" if stderr_tail else ""
        raise TranscriptionError(
            "VoxIntel could not preprocess this audio file with ffmpeg." + detail
        )

    return temp_path


def _run_ffmpeg_decode(ffmpeg_path: Path, source_path: Path, output_path: Path,
enhance_audio: bool) -> subprocess.CompletedProcess:
    command = [
        str(ffmpeg_path),
        "-y",
        "-i",
        str(source_path),
    ]
    if enhance_audio:
        command.extend(
            [
                "-af",
                "highpass=f=80,lowpass=f=3800,afftdn=nf=-20,dynaudnorm",

# ... [Truncated: 362 additional lines in production codebase] ...
```

voxintel/insights.py (TextRank Summarization & Action Item Extraction)

```
from __future__ import annotations

import re
from collections import Counter

import networkx as nx

from .ml import AISummaryUnavailable, generate_ai_summary
from .nlp import clean_sentences, clean_transcript, split_sentences, tokenize

ACTION_PATTERNS = [
    re.compile(
        r"(?i)^(?P<person>[A-Z][a-z]+(?:\s[A-Z][a-z]+)?)\s+(?:will|should|needs to|must|has
to|is going to)\s+(?P<task>.+?)(?:\s+(?:by|before|on)\s+(?P<deadline>[^.]+?))?[.]?$"
    ),
    re.compile(
        r"(?i)^(?:(?:once|after|as soon
as|when|regarding|before|if)\s+[^,;]+,\s*)?(?:i'll|i\s+will|we'll|we\s+will|i\s+can|we\s+can
|i\s+should|i\s+need\s+to|we\s+need\s+to|i\s+am\s+going\s+to|we\s+are\s+going\s+to)\s+(?P<ta
sk>.+?)(?:\s+(?:by|before|on|as soon as)\s+(?P<deadline>[^.]+?))?[.]?$"
    ),
    re.compile(
        r"(?i)^action item[:\-]?\s*(?P<person>[A-Z][a-z]+)?\s*(?:will|should|needs
to)?\s*(?P<task>.+?)(?:\s+(?:by|before|on)\s+(?P<deadline>[^.]+?))?[.]?$"
    ),
]

DECISION_PATTERN =
```

```
re.compile(r"(?i)\b(decide|decided|decision|agreed|approved|finalized|confirmed)\b")
DECISION_STRONG_PATTERN = re.compile(
    r"(?i)\b(decide|decided|decision|approved|agreed|finalized|confirmed|moving
forward|temporary workaround|workaround)\b"
)
DATE_PATTERN = re.compile(
    r"\b(?:\d{1,2}[/-]\d{1,2}[/-]\d{2,4}|"
    r"(?:jan|feb|mar|apr|may|jun|jul|aug|sep|oct|nov|dec)[a-z]*\s+\d{1,2}(?:,\s*\d{4})?)\b",
    re.IGNORECASE,
)
SPEAKER_PATTERN = re.compile(r"(?m)^\s*([A-Z][a-z]+(?:\s[A-Z][a-z]+)?)\s*:")
SPEAKER_INLINE_PATTERN = re.compile(r"\b([A-Z][a-z]+(?:\s[A-Z][a-z]+)?)\s*:")
FULL_NAME_PATTERN = re.compile(r"\b([A-Z][a-z]+ [A-Z][a-z]+)\b")
TOPIC_CUE_PATTERN = re.compile(

r"(?i)\b(issue|issues|bug|bugs|authentication|ui|design|release|timeline|deadline|demo|suppo
rt|testing|review|fix|plan|notes|documentation)\b"
)
DEADLINE_PATTERN = re.compile(
    r"(?i)\b(by tomorrow|tomorrow|next week|next month|this week|today|by [A-Z][a-
z]+day(?:\s+at\s+\d{1,2}(?::\d{2})?\s*(?:AM|PM))?|\d{1,2}(?::\d{2})?\s*(?:AM|PM)(?:\s+today)
?)\b"
)
WEAK_ACTION_PATTERN = re.compile(r"(?i)\b(we'?ll try|we'?ll ensure|sounds good|good
idea|good point|maybe|should discuss|let'?s discuss)\b")
VAGUE_DECISION_PATTERN = re.compile(r"(?i)\b(maybe|might|should|let'?s
discuss|perhaps|could)\b")
COMMON_FIRST_NAMES = {

"alex","amy","andrew","anna","asha","ben","chris","christy","daniel","david","dove","eric","
fabian",

"gabe","george","jessica","john","josh","karina","kenny","meera","paul","rahul","scott","sid
","virginia"
}


def infer_speaker_activity(transcript: str) -> list[dict[str, int]]:
    speaker_counts = Counter(SPEAKER_PATTERN.findall(transcript))
    return [{"speaker": speaker, "mentions": count} for speaker, count in
speaker_counts.most_common()]


def normalize_for_summary(text: str) -> str:
    normalized = re.sub(r"\s+", " ", text).strip()
    normalized = SPEAKER_INLINE_PATTERN.sub("", normalized)
    normalized = re.sub(r"\b(today|hello everyone|good morning team)\b[:,]?", "",
normalized, flags=re.IGNORECASE)
    normalized = re.sub(r"\s+", " ", normalized).strip(" ,")
    return normalized


def _sentence_similarity(sentence_a: str, sentence_b: str) -> float:
    tokens_a = set(tokenize(sentence_a))
    tokens_b = set(tokenize(sentence_b))
    if not tokens_a or not tokens_b:
        return 0.0
    overlap = len(tokens_a & tokens_b)
    return overlap / max((len(tokens_a) * len(tokens_b)) ** 0.5, 1.0)


def _textrank_scores(sentences: list[str]) -> dict[int, float]:
    if len(sentences) <= 1:
        return {0: 1.0} if sentences else {}

    graph = nx.Graph()
    for index in range(len(sentences)):
```

```
        graph.add_node(index)
    for left in range(len(sentences)):
        for right in range(left + 1, len(sentences)):
            similarity = _sentence_similarity(sentences[left], sentences[right])
            if similarity > 0:
                graph.add_edge(left, right, weight=similarity)

    if graph.number_of_edges() == 0:
        return {index: 1.0 for index in range(len(sentences))}
    return nx.pagerank(graph, weight="weight")


def _generate_rule_based_summary(sentences: list[str], tokens: list[str]) -> str:
    if not sentences:

# ... [Truncated: 369 additional lines in production codebase] ...
```

voxintel/ollama_polish.py (Local Ollama Third-Person MoM Synthesis)

```
from __future__ import annotations

import json
import os
import re
from pathlib import Path
from urllib import request, error

from .exports import render_mom_html, render_mom_pdf
from .optimized_pipeline import OptimizedPipelineError, run_optimized_voxintel

DEFAULT_OLLAMA_HOST = "http://127.0.0.1:11434"
DEFAULT_OLLAMA_MODEL = "llama3.2:3b"


class OllamaPolishError(RuntimeError):
    """Raised when local Ollama polishing cannot continue."""


def _get_ollama_settings() -> tuple[str, str]:
    host = os.getenv("OLLAMA_HOST", DEFAULT_OLLAMA_HOST).rstrip("/")
    env_model = os.getenv("OLLAMA_MODEL", "").strip()
    if env_model:
        return host, env_model

    try:
        req = request.Request(f"{host}/api/tags", headers={"Content-Type":
"application/json"}, method="GET")
        with request.urlopen(req, timeout=1.5) as resp:
            data = json.loads(resp.read().decode("utf-8"))
            installed = [m.get("name", "") for m in data.get("models", [])]
            for candidate in ["llama3.2:3b", "llama3.2", "gemma3:1b", "gemma3", "gemma2",
"qwen2.5:0.5b", "qwen2.5", "phi3", "phi:latest", "mistral"]:
                for inst in installed:
                    if candidate in inst:
                        return host, inst
            if installed:
                return host, installed[0]
    except Exception:
        pass

    return host, DEFAULT_OLLAMA_MODEL
```

```
def resolve_speaker_mapping(segments: list[dict], attendees: list[str] | None = None) ->
dict[str, str]:
    """Map anonymous Speaker clusters (Speaker 1, 2...) to real attendee names using
conversational evidence."""
    if not attendees:
        return {}

    mapping: dict[str, str] = {}
    assigned_attendees: set[str] = set()
    clusters = sorted(list(set(s.get("speaker", "") for s in segments if s.get("speaker",
"").startswith("Speaker"))))

    # 1. Direct self-introductions ("This is [Name]", "I am [Name]")
    for att in attendees:
        att_lower = att.lower()
        for s in segments:
            text_lower = (s.get("text") or "").lower()
            spk = s.get("speaker")
            if not spk or spk in mapping:
                continue
            if f"this is {att_lower}" in text_lower or f"i am {att_lower}" in text_lower
or f"i'm {att_lower}" in text_lower:
                mapping[spk] = att
                assigned_attendees.add(att)
                break

    # 2. Conversational Adjacency Pairs:
    # A) "Thanks, [Name]" / "Thank you, [Name]" -> the PRECEDING turn belonged to [Name]
    for i, s in enumerate(segments):
        if i == 0:
            continue
        txt = (s.get("text") or "").lower()
        prev_spk = segments[i - 1].get("speaker")
        if not prev_spk or prev_spk in mapping:
            continue
        for att in attendees:
            if att in assigned_attendees:
                continue
            if re.search(rf"\b(?:thanks|thank you),?\s+{att.lower()}\b", txt):
                mapping[prev_spk] = att
                assigned_attendees.add(att)
                break

    # B) "[Name], what/how/can/could/is/are/do/did...?" -> the NEXT turn answering belongs
to [Name]
    for i, s in enumerate(segments[:-1]):
        txt = (s.get("text") or "").lower()
        next_s = segments[i + 1]
        next_spk = next_s.get("speaker")
        if not next_spk or next_spk in mapping:
            continue
        next_txt = (next_s.get("text") or "").lower()
        for att in attendees:

# ... [Truncated: 577 additional lines in production codebase] ...
```